JAKE SONG

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APPOINTMENTS

Stanford University, CA, USA June 2023-
NIH Postdoctoral Fellow in Mechanical Engineering
Advisor: Ovijit Chaudhuri

EDUCATION

Massachusetts Institute of Technology, MA, USA May 2023
Ph.D. in Materials Science and Engineering & Program in Polymer and Soft Matter
Advisors: Gareth H. McKinley and Niels Holten-Andersen

Northwestern University, IL, USA May 2017
B.S. in Materials Science and Engineering
Advisor: Sinan Keten

HONORS AND AWARDS

NIH F32 Kirschstein-NRSA Postdoctoral Fellowship	2024-2027
Rising Star in Materials Science and Engineering	2026
Rising Star in Soft and Biological Matter	2025
Rising Star in Engineering in Health	2025
ACS PMSE Future Faculty	2025
USNC/TAM Thomas J. R. Hughes Fellowship	2024
APS DSOFT Emerging Soft Matter Excellence Award Finalist	2023
APS DPOLY Frank J. Padden Jr. Award Finalist	2022
MIT MathWorks Graduate Fellowships	2021-2022 and 2022-2023
MIT Mechanical Engineering Research Exhibition, First Place	2021
MIT Lemelson-Vest Award	2021
MIT Polymer Day Symposium, Second Place	2018 and 2019
MIT Robert M. Rose Presidential Fellowship	2017
Northwestern University Harold B. Gotaas Award	2017
Northwestern University John E. Hilliard Award	2017

PUBLICATIONS

(*Equal author contribution) ([#]Corresponding author)

1. **Song, J.,** Loveland, S., Yang, F., Bhutani, N., Levenston, M. E., & Chaudhuri, O. "Hydrogel Confinement of the Secreted Nascent Matrix Imposes Solid Stress and Mediates Pathology of Chondrocytes". *bioRxiv.* (2026) https://doi.org/10.64898/2026.10.19.760217
2. Lau, R., Zhu, J., Dehghany, M., Slyman, R., **Song, J.,** Shenoy, V., & Chaudhuri, O. "Matrix Yielding Drives Symmetry Breaking of Growing Tumor Spheroids in Confining Matrices". *Under Review.*
3. Loveland, S., Tong X., Ayushman, M., Lee, H-P., Johannsen, J. M., Weber, C., **Song, J.,** Tai, M., Chaib, Y., Chaudhuri, O., & Yang. F. "Adaptable Sliding Hydrogels Enable Pericellular Pocket Formation while Enhancing MSC Chondrogenesis and Survival in 3D". *Bioactive Materials.* (2026)
4. **Song, J.[#],** Deiss-Yehiely, E., Yesilata, S., & McKinley, G. H. "Strain Stiffening Universality in Composite Hydrogels and Soft Tissues". *Nature Physics.* (2025)

5. Degen, G. D., Stevens, C. A., Cárcamo-Oyarce, G., **Song, J**., Bej, R., Tang, P., Haag, R., Ribbeck, K. R. & McKinley, G. H. "Mussel-Inspired Cross-Linking Mechanisms Enhance Gelation and Adhesion of Multifunctional Mucin-Derived Hydrogels". *Proceedings of the National Academy of Sciences.* (2025)
6. **Song, J.**[#] "Non-linear Rheology of Melted Cheddar Cheese". *Journal of Food Engineering.* (2025)
7. Le Roy, H.*, **Song, J.*,** Lundberg, D., Zhukhovitskiy, A. V., Johnson, J. A., McKinley, G. H., Holten-Andersen, N., & Lenz, M. "Valency Can Control the Non-Exponential Viscoelastic Relaxation in Multivalent Reversible Gels". *Science Advances.* (2024)
8. **Song, J.,** Kim, S., Saouaf, O., Owens, C., McKinley, G. H., & Holten-Andersen, N. "Soft Viscoelastic Magnetic Hydrogels from the In Situ Mineralization of Iron Oxide in Metal-Coordinate Polymer Networks". *ACS Applied Materials and Interfaces.* (2023)
9. **Song, J.,** Holten-Andersen, N., & McKinley, G. H., "Non-Maxwellian Viscoelastic Stress Relaxations in Soft Matter". *Soft Matter.* (2023)
10. **Song, J.*,** Khare, E*., Rao, L., & Holten-Andersen, N. "Coordination Stoichiometry Effects on the Binding Hierarchy of Histamine and Imidazole-M^{2+} Complexes". *Macromolecular Rapid Communcations.* (2023)
11. Khare, E., Cazzell., S., **Song, J**., Holten-Andersen, N., & Buehler, M. "Molecular Understanding of Ni^{2+}-Nitrogen Family Metal-Coordinated Hydrogel Relaxation Times Using Free Energy Landscapes." *Proceedings of the National Academy of Sciences.* (2023)
12. **Song, J.,** Zhang, Q., de Quesada, F., Rizvi, M. H., Ilavsky, J., Tracy, J. B., Narayanan, S., Del Gado, E., Leheny. R. L., Holten-Andersen, N., & McKinley, G. H. "Microscopic Dynamics Underlying the Stress Relaxation of Arrested Soft Materials." *Proceedings of the National Academy of Sciences.* (2022)
13. **Song, J.*,** Li, Q.*, Chen, P., Keshavarz, B., Chapman, B., Tracy, J. B., McKinley, G. H., & Holten-Andersen, N. "Dynamics of Dual Metal-Coordinate Networks with Ion and Nanoparticle Cross-Link Junctions." *Journal of Rheology.* (2022)
14. Kim, S., Registsky, A. U., **Song, J.,** Ilavsky, J., McKinley, G. H., & Holten-Andersen, N. "*In-Situ* Mechanical Reinforcement of Polymer Hydrogels via Metal-Coordination Crosslink Mineralization." *Nature Communications.* (2021)
15. Jia, F., **Song, J.,** Kubiak, J., Santos, P. J., Sano, K., Holten-Andersen, N., Zhang, K., & Macfarlane, R. J. "Brush Polymers as Nanoscale Building Blocks for Hydrogel Synthesis." *Chemistry of Materials.* (2021)
16. **Song, J.,** Rizvi, M. H., Lynch, B. B., Ilavsky, J., Mankus, D., Tracy, J, B., McKinley, G. H., & Holten-Andersen, N. "Programmable Anisotropy and Percolation in Supramolecular Patchy Particle Gels." *ACS Nano.* (2020)
17. Lin, X., Liu, X., Liu, J., Yuk, H., Loh, H., Parada, G. A., Settens, C., **Song, J.,** Masic, A., McKinley, G. H., & Zhao, X. "Anti-Fatigue-Fracture Hydrogels." *Science Advances.* (2019)
18. **Song, J.*,** Kahraman, R.*, Collinson, D.*, Xia, W., Brinson, L. C., & Keten, S. "Temperature Effects on the Nanoindentation Characterization of Stiffness Gradients in Confined Polymers". *Soft Matter.* (2019)
19. **Song, J.,** Hsu, D. D., Shull, K. R., Phelan Jr, F. R., Douglas, J. F., Xia, W., & Keten, S. "Coarse-Grained Modelling of Polymer Viscoelasticity via Energy Renormalization." *Macromolecules.* (2018)
20. Xia, W., **Song, J.,** Krishnamurthy, N., Phelan Jr, F. R., Keten, S., & Douglas, J. F. "Energy Renormalization for Coarse-Graining the Dynamics of a Model Glass-Forming Liquid". *Journal of Physical Chemistry B.* (2018)
21. Xia, W.*, **Song, J.*,** Jeong, C., Hsu, D. D., Phelan, F. R., Douglas, J. F., & Keten, S. "Energy Renormalization for Temperature Transferable Coarse-Graining of Polymer Dynamics". *Macromolecules.* (2017)
22. Hsu, D. D., Xia, W., **Song, J.,** & Keten, S. "Dynamics of Interacting Interfaces in Polymer Bilayer Thin Films". *MRS Communications.* (2017)
23. Xia, W., **Song, J.,** Hsu, D. D., & Keten, S. "Side-Group Size Effects on Interfaces and Glass Formation in Supported Polymer Thin Films" *The Journal of Chemical Physics.* (2017)

24. Xia, W.*, **Song, J.*,** Hsu, D. D., & Keten, S. "Understanding the Interfacial Mechanical Response of Nanoscale Polymer Thin Films via Nanoindentation". *Macromolecules.* (2016)
25. Xia W., **Song, J**. Meng, Z., & Keten, S. "Designing Multi-Layer Graphene-Based Assemblies for Enhanced Toughness in Nacre-Inspired Nanocomposites". *Molecular Systems Design & Engineering.* (2016)
26. Hsu, D. D., Xia, W., **Song, J.,** & Keten, S. "Glass-Transition and Side-Chain Dynamics in Thin Films: Explaining Dissimilar Free Surface Effects for Polystyrene vs Poly(methyl methacrylate)". *ACS Macro Letters* (2016)

TEACHING EXPERIENCE

Guest lecturer for 20.463: Biomaterials Science and Engineering (MIT) 2022
Teaching assistant for 3.034: Organic and Biomaterials Chemistry (MIT) 2020
Teaching assistant for MechE 418: Multiscale Modelling and Simulation in Fluid Mechanics (Northwestern) 2017

MENTORSHIP EXPERIENCE

Aarushi Kumar, on solid stress in the articular cartilage (Stanford) 2025
Sophia Sonnert, on platelet interactions with fibrin-based hydrogels (Stanford) 2024
Shaleen Thiengmany, on loading device design for confocal imaging of hydrogel plasticity (Stanford) 2024
Syahmi Ali Mohd Rizal, on imaging cell and hydrogel plasticity under external loading (Stanford) 2024
Luan Dang, on machine learning characterization of biopolymer gel plasticity (Stanford) 2024
Adam Benmira, on loading device design for confocal imaging of hydrogel plasticity (Stanford) 2023
Uwizerwa Sonia Karita, on elastic energy storage in elastomeric materials (Stanford) 2023
Marcelo Garcia, on tension-compression asymmetry of fracture toughness in soft materials (MIT) 2022
Serra Yesilata, on compression stiffening in tissue-mimetic materials (MIT) 2022
Omar Laris, on in situ mineralization of nanoparticles in metal-coordinate thermoplastics (MIT) 2020
Carolyn Will, on rheometric measurement of torsional fracture toughness for soft materials (MIT) 2019
Helen Read, on rheometric measurement of torsional fracture toughness for soft materials (MIT) 2018
Olivia Saouaf, on in situ mineralization of magnetite in polyphenolic hydrogels (MIT) 2018
Felipe de Quesada, on chemo-mechanical design of nanoparticle-coordinated hydrogels (MIT) 2018
Danielle Herman, on colorimetric analysis of histidine-metal coordination interactions (MIT) 2018
Guobiao Li, on coarse-grained modelling high-frequency viscoelasticity of polybutadiene (Northwestern) 2017

SERVICE EXPERIENCE

Executive board member and postdoc representative of the Stanford Polymer Collective 2025-2026
Research mentor for first-generation low-income students as part of the Stanford Summer Fellows Program 2024
Session chair for the Colloidal Gels session at the march meeting of the American Chemical Society 2022
Volunteer for various science outreach initiatives hosted by Stanford, MIT, and the Society of Rheology 2017-
Peer-reviewer for Nature (2025), Science Advances (2024), Polymer Chemistry (2023), ACS Macro Letters (2019), Macromolecules (2017), and Journal of Chemical Physics (2017)

INDUSTRY EXPERIENCE

Motif FoodWorks, MA, USA 2021
PhD Summer Intern, R&D – developed rheological protocol for fingerprinting the viscoelasticity of dairy cheeses, and designed plant-based cheese substitutes using zein protein aggregates